Model Selection Justification for the NAVIG Geolocalization Pipeline

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Contents

1 Overview

NAVIG [Yu et al., 2025] is a multi-stage vision-language geolocalization pipeline. Given a geo-tagged photograph, the system produces an estimated GPS coordinate—expressed as (latitude, longitude, country, city)—by chaining six stages: (1) macro-level scene reasoning, (2) visual grounding of salient objects via GroundingDINO [Liu et al., 2024c], (3) retrieval-augmented lookup against a geographic guidebook indexed with CLIP [Radford et al., 2021] and FAISS [Johnson et al., 2021], (4) VLM-based commentary on cropped patches, (5) OCR-driven OpenStreetMap/Nominatim search, and (6) final coordinate synthesis. This document justifies the selection and configuration of every vision-language model (VLM) used in these stages and reports an empirical evaluation that extends the original paper in four directions the paper did not cover.

Why this evaluation extends the original paper. The original NAVIG paper evaluated three 7B-parameter SFT-trained models (MiniCPM-V-2.6, LLaVA-1.6-Vicuna-7B, Qwen2-VL-7B) and reported their performance on the GWS5k benchmark as the primary result, with im2gps3k results in the appendix. The best result on im2gps3k was Navig-Qwen2-VL at GeoScore 3,482. The paper's ablation study removed entire pipeline modules (Reasoner, Searcher) but did not isolate the contribution of individual evidence types (RAG vs. patch commentary vs. OSM). No experiment varied which model handled which stage: all three tested models ran the full pipeline with their own SFT adapter, and no controlled Stage-6 swap was performed. The paper identified four limitations—small training corpus, street-view domain constraint, 7B model size ceiling, and pipeline component conflicts—but provided no empirical diagnosis of which limitation most limits performance.

This evaluation addresses those gaps directly:

1. **Evidence attribution:** A per-record analysis separates the contribution of RAG, patch commentary (COMMENT), and OSM evidence at the sample level—the first such decomposition for NAVIG. The finding that RAG *systematically hurts* performance (-356 to -756 GeoScore points across all models) directly contradicts the original paper's implicit assumption that all Searcher components contribute positively.
2. **Larger-model scaling:** LLaMA-3.2-11B is the first 11B-parameter model evaluated in the NAVIG pipeline. Its failure-excluded GeoScore of 3,357 approaches the original paper's best result (Qwen2-VL SFT: 3,482), achieved without any NaviClues fine-tuning—a result with strong implications for whether SFT or raw model capacity is the primary performance driver.
3. **Stage-level bottleneck isolation:** The Stage-6 swap experiment (Section 7) is the first controlled test of whether Stage 6 synthesis quality or Stage 1 reasoning quality is the performance bottleneck. This experiment is not present in any form in the original paper.
4. **Geographic error analysis:** A regional breakdown (Section 10) reveals a 47% GeoScore gap between European images (3,205) and African images (1,686), quantifying the geographic bias that the original paper acknowledged as a qualitative limitation but did not measure.

Our evaluation also replicates the original paper's LLaVA result closely (2,626 vs. the paper's 2,592), confirming that the evaluation setup is consistent with the original. Qwen2-VL—the original paper's strongest model—failed entirely in our implementation due to a prompt-format incompatibility (Section 10.10) that is diagnosed here and represents a concrete engineering target.

Three distinct experimental conditions are used across the model runs:

1. **Full pipeline with SFT** (primary models) — a single model family drives both Stage 1 reasoning (via a LoRA-fine-tuned SFT variant [Hu et al., 2022]) and Stages 4–6 (via the same checkpoint loaded without the adapter). SFT is applied *only* at Stage 1; Stages 4–6 always use the pretrained base weights.
2. **Full pipeline, zero-shot** (comparison models) — a model runs all six stages without any NAVIG-specific fine-tuning, providing zero-shot baselines that isolate the benefit of Stage 1 SFT.
3. **Stage-6 swap only** — the pre-computed Stage-1–5 outputs from the LLaVA full-pipeline run are held fixed, and a different model is substituted exclusively at Stage 6. This tests whether coordinate synthesis quality is the primary bottleneck, independent of upstream evidence generation.

Table 1 summarises all seven model configurations and their experimental roles.

2 Evaluation Dataset: im2gps3k

All experiments in this document are evaluated on the **im2gps3k** benchmark, introduced by Vo et al. [2017] as a harder, larger complement to the original 237-image IM2GPS test set of Hays and Efros [2008]. The dataset consists of 2,997 GPS-tagged photographs sourced from Flickr—the same community photograph platform used to construct the underlying IM2GPS retrieval database—but without the manual curation applied to the original test set. This deliberate absence of filtering means im2gps3k contains the full range of image types submitted by real photographers: recognisable landmarks, urban street scenes, rural and natural landscapes, wildlife photographs, and close-up subject photography. The challenge of geo-localization on this benchmark reflects the challenge of geo-localization in the real world.

Geographic distribution. As shown in Figure 1, the dataset is strongly biased toward Western-hemisphere photographers: Europe accounts for 35.6% (1,067 images) and North America for 31.2% (936 images), together representing two-thirds of the benchmark. Asia contributes 15.2% (455 images), while Africa (5.3%), South America (3.8%), and Oceania (1.4%) are heavily under-represented relative to their landmass or population. A further 7.5% of images do not fall within the defined continental bins (polar regions, island chains, or ambiguous coordinates). This distribution directly mirrors the geographic bias of Flickr's userbase in the benchmark's collection window, and it has two important consequences for evaluation: (1) models will see more training-distribution-similar content in European and North American test images, inflating aggregate metrics; and (2) reported performance on under-represented regions is computed over small sample sizes and therefore has high variance.

Image content and difficulty. The benchmark spans at least four qualitatively distinct image categories, each presenting a different failure mode for the NAVIG pipeline:

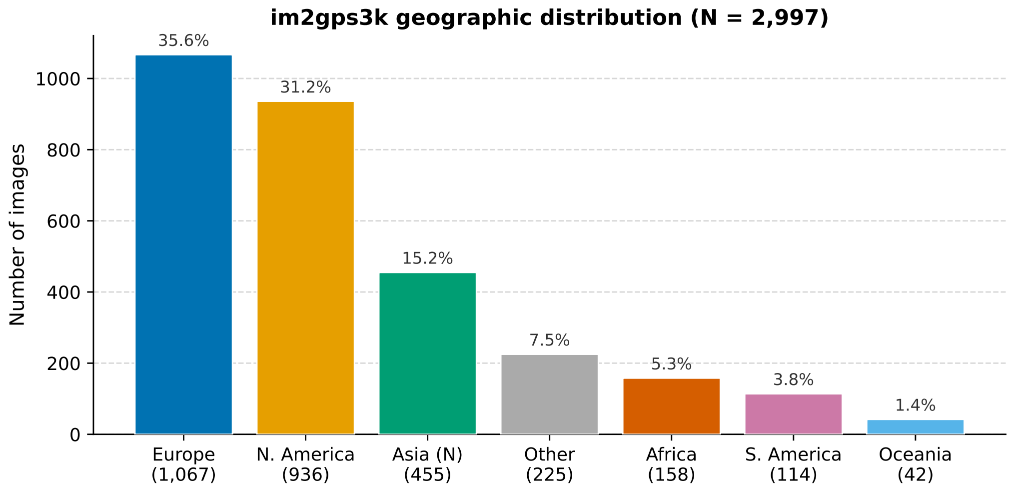

Figure 1: Geographic distribution of the 2,997-image im2gps3k benchmark by broad continental region. Europe and North America together account for two-thirds of the dataset, reflecting the geographic bias of Flickr photographers in the collection window.

1. **Landmark photographs.** Images of globally recognisable structures (the Eiffel Tower, Trafalgar Square, the Taj Mahal) can be localised to within metres by VLMs that have learned to associate these structures with specific GPS coordinates from pretraining data. These images are trivially easy and inflate aggregate accuracy at fine-grained thresholds.
2. **Text-bearing urban scenes.** Street scenes containing legible local text—shop signs, road signs, street name plates—can be localised by the OCR+Nominatim chain in Stage 5, provided the text is in a language or script with distinctive geographic specificity (Arabic, Cyrillic, Thai, etc.).
3. **Generic urban infrastructure.** Parking lots, plain building facades, and undistinctive roadways carry subtle visual cues—bollard colours, road marking styles, kerb design, licence plate proportions—that human experts (e.g. GeoGuessr players) can read but current VLMs frequently misinterpret. These images produce the hemisphere-flipping errors visible in Figure 2 (left, bottom row): the model correctly identifies a temperate-climate English-speaking country but places it on the wrong side of the Earth.
4. **Natural landscapes and subject photography.** Images showing wildlife, close-up flora, or featureless natural terrain provide little or no geographic signal for a VLM operating at the text-prompt level. The pipeline can only hallucinate a species-based geographic prior (e.g. associating a crocodile photograph with Cuba rather than Thailand) or default to a polar confusion when confronted with a barren volcanic landscape. No amount of upstream SFT or downstream RAG retrieval can recover meaningful localisation from an image that contains no geographic information.

Figure 2 illustrates the contrast between categories 1 and 4 concretely, comparing three near-perfect predictions against three maximum-error failures from the LLaVA-1.6 full pipeline. The failures are not random: they are predictable from the image content alone. This is an important caveat when interpreting aggregate GeoScore numbers—a model that achieves GeoScore 2626 is not "26% of optimal" uniformly; it is very good at landmark images and completely uninformative on a systematic subset of the benchmark.

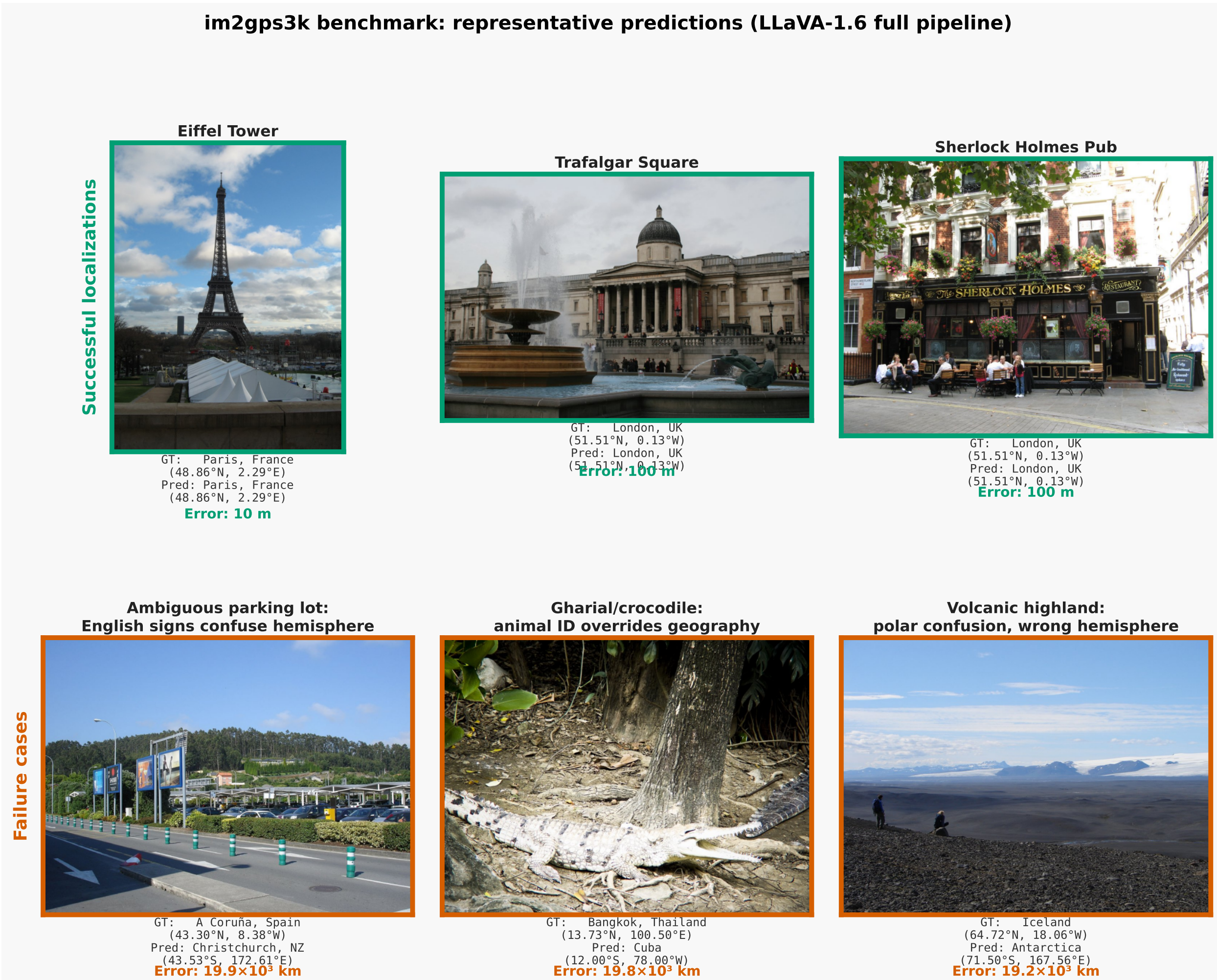

Figure 2: Representative im2gps3k predictions from the LLaVA-1.6 full pipeline. **Top row (successes):** Landmark-rich images are localised to within metres. Left: Eiffel Tower (Paris), 0.01 km error. Centre: National Gallery, Trafalgar Square (London), 0.10 km error. Right: The Sherlock Holmes pub, Northumberland Street (London), 0.10 km error—the pub name and street sign are both legible, making it trivially identifiable. **Bottom row (failures):** Images providing little or no geographic signal produce antipodal errors exceeding 19,000 km. Left: A commercial parking area in A Coruña, Spain; English-language signage and temperate greenery caused the model to predict Christchurch, New Zealand (19,931 km error). Centre: A gharial photographed in Bangkok, Thailand; the model associated the species with Cuba (19,763 km error). Right: The Icelandic highland near Mýrdalsjokull; the barren volcanic terrain and distant glacier were interpreted as Antarctica (19,226 km error). The failure cases are categorically different from the successes—they contain almost no unambiguous geographic signal—and represent a systematic, not random, source of error.

Suitability for NAVIG evaluation. im2gps3k was selected as the primary evaluation benchmark for three reasons. First, the original NAVIG paper [Yu et al., 2025] reported results on im2gps3k, making it the natural choice for assessing comparability with prior work. Second, at 2,997 images it is large enough to support statistically meaningful breakdown by geographic region and evidence type, as reported in Section 10. Third, the benchmark is old enough (images collected before ∼2017) that it is unlikely to have been included in the pretraining corpora of the VLMs evaluated here, reducing the risk of inadvertent memorisation.

A known limitation is that im2gps3k over-represents landmark and tourist imagery relative to the realistic distribution of geo-localization queries in deployment. This inflates reported accuracy at fine-grained distance thresholds (1 km and 25 km) relative to what a deployed system would achieve on arbitrary queries. Future work should include evaluation on GWS15K, which has a more uniform geographic distribution, and on street-level datasets closer in character to GeoGuessr gameplay—the domain on which the Stage 1 models were fine-tuned.

3 Training Dataset: NaviClues

The Stage 1 reasoning models are fine-tuned on **NaviClues**, a proprietary dataset of geolocalization gameplay transcripts collected from expert GeoGuessr players and paired with the photographs they were shown. Each record stores an image path, a YouTube gameplay URL, the GeoGuessr challenge URL, the ground-truth GPS coordinate, a timestamped player commentary transcript, the GeoScore awarded (0–5000 points on the standard exponential scale), and the Haversine error in kilometres. Images are hosted publicly on Hugging Face at `huggingCode11/NAVICLUES`; LoRA adapters are at `huggingCode11/NAVIG`.

The raw corpus contains 2,637 samples, filtered to 1,638 samples after quality and diversity screening (`filtered_data.jsonl`), and further refined to 390 high-quality samples (`quality_data.jsonl`) used for the final SFT runs. This aggressive 85% reduction reflects the difficulty of extracting clean training signal from unstructured gameplay video: many transcripts are incomplete, redundant, or contain irrelevant commentary. The 390 retained samples are those where the player's verbal reasoning is geographically specific, internally consistent, and achieves a GeoScore above a quality threshold.

The GeoScore metric used for both training-data selection and evaluation follows the standard exponential decay formula from the original NAVIG paper:

$$\mathrm{GeoScore}(d) = 5000 \cdot \exp\Big(-\frac{d}{1492.7}\Big), \tag{1}$$

where d is the Haversine distance in kilometres between the predicted and ground-truth coordinates. A score of 5000 indicates a perfect prediction; a score of 0 corresponds to a prediction on the opposite side of the globe ($d \gtrsim 10{,}000\,\mathrm{km}$).

NaviClues vs. im2gps3k: domain mismatch. NaviClues images derive from GeoGuessr gameplay, which uses Google Street View panoramas—near-ground-level, 360-degree imagery collected by dedicated camera rigs under standardised exposure. im2gps3k images are amateur Flickr photographs with arbitrary composition, exposure, and subject matter. The Stage 1 model is therefore fine-tuned on a somewhat different visual domain than the one it is evaluated on. This domain gap is one reason why the SFT benefit—while real—is more modest than might be expected from the quality of the training transcripts. Closing this gap, by augmenting NaviClues with Flickr-style imagery or by fine-tuning directly on a sample of im2gps3k images, is a concrete avenue for future improvement.

4 Fine-Tuning and Inference Infrastructure

All SFT runs use the **ms-Swift** framework [ModelScope Community, 2024] (version 2.5.0.post1) with LoRA adapters [Hu et al., 2022] of rank 16. Training converges at step 534 for each model family, producing the shared `checkpoint-534` naming convention visible in the `vlms/` directory. At inference time, the `_sft` variant loads the pretrained weights and then applies the saved LoRA checkpoint via `Swift.from_pretrained(..., inference_mode=True)`.

The pipeline runs on single NVIDIA RTX A6000 GPUs (48 GB VRAM, compute capability SM 8.6) under CUDA 12.1 and PyTorch 2.4.0. All models are loaded in `float16` (except InternVL2-8B, which uses `bfloat16` for numerical stability as recommended by the model authors). To avoid holding two copies of a 7 B model in VRAM simultaneously, the reasoning model (Stage 1) is loaded, used, and freed before the base model (Stages 4–6) is instantiated. The SFT model always uses the ms-Swift inference path regardless of runtime flags, because vLLM's multimodal LoRA support for LLaVA-NeXT is not yet stable.

Optional vLLM [Kwon et al., 2023] acceleration (version 0.5.5) is available for Stages 4–6 via the `--use_vllm` flag. vLLM applies *only* to the base model in these later stages; Stage 1 always falls back to ms-Swift.

For visual grounding (Stage 2), NAVIG uses GroundingDINO [Liu et al., 2024c] with a Swin-T backbone (`groundingdino_swint_ogc.pth`), configured by default with box threshold 0.65 and text threshold 0.55, detecting three object categories: road signs, house facades, and building signs. Stage 3 retrieval is built on CLIP ViT-B/32 [Radford et al., 2021] embeddings over a curated geographic guidebook, stored in a FAISS [Johnson et al., 2021] L2 index; the distance threshold for including a retrieved clue in the Stage 6 prompt is set to 30 (L2 units in the embedding space).

5 Primary Pipeline Models (with Supervised Fine-Tuning)

The three models below are the core subjects of NAVIG's supervised fine-tuning study. Each was fine-tuned on the 390-sample NaviClues quality subset using LoRA adapters managed by ms-Swift, as described in Section 4.

5.1 LLaVA-1.6-Vicuna-7B (`LLaVA` / `LLaVA_sft`)

Architecture. LLaVA-NeXT (version 1.6) [Liu et al., 2024b], which builds on LLaVA-1.5 [Liu et al., 2024a], couples a CLIP ViT-L/14@336px vision encoder [Radford et al., 2021] with Vicuna-7B [Chiang et al., 2023] as the language backbone. Images are tiled into up to six high-resolution patches (a 2×2 grid plus one global thumbnail) before being projected into the language embedding space, giving the model substantially finer spatial resolution than the LLaVA-1.5 family. The ms-Swift model type is `llava1_6-vicuna-7b-instruct`; the LLaVA-NeXT processor requires the `patch_size` and `vision_feature_select_strategy` attributes to be set explicitly (patched in `llm.py::_fix_llava_next_processor` to avoid an upstream deprecation warning in transformers 4.45.2).

Why chosen. LLaVA-1.6-Vicuna-7B served as the original baseline model in the first published iteration of NAVIG [Yu et al., 2025]. Its inclusion is therefore both a continuity decision—preserving direct comparability with prior published numbers—and an architectural one: the tile-based high-resolution encoding is well-suited to street-level images containing small but legible signs and text. At 7 B parameters the model fits comfortably within a single 48 GB A6000 GPU in `float16` (≈14 GB VRAM).

Fine-tuning. A LoRA adapter (rank 16) was trained on the NaviClues quality subset and is stored at `vlms/NAVIG/llava1_6-vicuna-7b-instruct/checkpoint-534`. The SFT variant (`LLaVA_sft`) is used *only* in Stage 1 to generate the macro geographic reasoning chain. The base model (`LLaVA`) handles Stages 4–6 without the adapter.

vLLM path. An optional `LLaVA_vllm` / `LLaVA_sft_vllm` path accelerates throughput via vLLM with LoRA pass-through. Because vLLM's multimodal LoRA support for LLaVA-NeXT is not fully stable, vLLM is applied *only* to the base model when `--use_vllm` is set; Stage 1 always uses the ms-Swift path.

5.2 Qwen2-VL-7B-Instruct (`Qwen` / `Qwen_sft`)

Architecture. Qwen2-VL-7B [Wang et al., 2024] is Alibaba's second-generation visual language model. It introduces a *Naive Dynamic Resolution* mechanism that allows the vision encoder to process images at their native aspect ratio and resolution rather than forcing a fixed square crop, and a *Multimodal Rotary Position Embedding* (M-RoPE) that jointly encodes spatial and temporal positions across the image and text sequence. The language backbone is Qwen2-7B. The ms-Swift model type is `qwen2-vl-7b-instruct`; weights from the subsequent Qwen2.5-VL release (which shares the same architecture and tokeniser) are forward-compatible with this model type in ms-Swift.

Why chosen. Qwen2-VL-7B achieves strong performance on OCR and document understanding benchmarks (e.g., TextVQA, OCRBench, DocVQA) relative to comparably sized models [Wang et al., 2024]. This is directly relevant to Stages 4–5 of NAVIG, which rely on reading small cropped text patches—signs, storefronts, road markings—to generate Nominatim search queries. Qwen2-VL's native-resolution encoding is particularly advantageous when patches are non-square or of variable aspect ratio, as is typical of GroundingDINO detections.

Fine-tuning. A LoRA adapter was trained under the same procedure as LLaVA and is stored at `vlms/qwen/checkpoint-534`. The `Qwen_sft` variant drives Stage 1; `Qwen` (base) drives Stages 4–6.

5.3 MiniCPM-V-2.6 (`CPM` / `CPM_sft`)

Architecture. MiniCPM-V-2.6 [Yao et al., 2024] is a compact multimodal model from ModelBest/OpenBMB. It is built around the SigLIP-400M vision encoder [Zhai et al., 2023] and the Qwen2-7B language model, totalling approximately 8 B parameters. It employs an adaptive visual encoding scheme and an efficient *token compression* mechanism that reduces the number of visual tokens passed to the language model, enabling longer effective context at lower memory cost. MiniCPM-V-2.6 additionally supports native multi-image input in a single forward pass. The ms-Swift model type is `minicpm-v-v2_6-chat`.

Why chosen. MiniCPM-V-2.6 offers a different architecture trade-off than LLaVA and Qwen: aggressive visual token compression reduces VRAM pressure during Stage 4's multi-patch commenting loop, and the native multi-image interface is directly relevant to stages that process batches of crops. Including MiniCPM tests whether the visual understanding bottleneck in NAVIG lies in image encoding quality (favouring Qwen or LLaVA) or in token-efficient summarisation (favouring MiniCPM). The SigLIP-400M encoder is trained with a sigmoid loss rather than a contrastive softmax, which empirically improves performance at smaller batch sizes [Zhai et al., 2023].

Fine-tuning. A LoRA adapter was trained identically to the other primary models and is stored at `vlms/cpm/checkpoint-534`. The `CPM_sft` variant drives Stage 1; `CPM` drives Stages 4–6.

6 Full-Pipeline Comparison Models (Zero-Shot, No SFT)

The three models in this section were each run through the *complete* six-stage NAVIG pipeline—all stages 1–6—without any NAVIG-specific fine-tuning. Each model therefore generated its own Stage 1 geographic reasoning chain, performed its own Stage 4 patch commentary, and synthesised its own Stage 6 coordinate prediction, all zero-shot from pretrained base weights. This design isolates the benefit of the Stage 1 LoRA adapter: comparing these zero-shot full-pipeline runs against the SFT-trained primary models (Section 5) directly measures how much NaviClues fine-tuning improves end-to-end accuracy when the same model family is held constant.

Each model was selected to vary a distinct architectural axis relative to the primary models: parameter scale (7B vs. 11B), visual encoding strategy (tile-based vs. dual-encoder), language backbone, and pretraining data distribution. A fourth comparison model, InternVL2-8B (implemented in `llm.py` and described in Section 6.4), was planned under the same design but has not yet been evaluated due to scheduling constraints.

Relationship to the Stage-6 swap experiment. Separately from the full-pipeline comparison runs, a *Stage-6 swap* experiment was conducted using `guess_only.py` (Section 7). In that experiment, the pre-computed Stage 1–5 outputs from the LLaVA full-pipeline run are held fixed, and LLaMA-3.2-11B-Vision-Instruct is substituted at Stage 6 only. LLaMA-3.2 is therefore the only model evaluated in both conditions—as a full-pipeline model generating its own upstream evidence, and as a Stage 6 guesser operating on LLaVA's upstream evidence. The comparison between these two conditions (Sections 10 and 10.10) directly tests whether the upstream reasoning chain or Stage 6 synthesis quality is the performance bottleneck.

6.1 Llama-3.2-11B-Vision-Instruct (`Llama32Vision`)

Architecture. Meta's Llama-3.2-11B-Vision-Instruct [Meta AI, 2024] grafts a cross-attention vision adapter onto a Llama 3.1 text backbone, scaling the combined model to 11 B parameters. The cross-attention fusion mechanism integrates visual features at multiple decoder layers rather than projecting them solely at the input, and the model was instruction-tuned specifically for visual question answering and visual reasoning tasks. The ms-Swift model type is `llama3_2-11b-vision-instruct`.

Why chosen. At 11 B parameters, LLaMA-3.2-11B is the largest model in the comparison set. It is hypothesised to have the strongest geographic commonsense reasoning capability, owing to the larger Llama 3.1 pretraining corpus [Meta AI, 2024], and is accordingly recommended as the default zero-shot model in the experimental workflow (see `guess_only.py`). At `float16` it requires approximately 22 GB of VRAM, fitting comfortably on a single 48 GB A6000 card.

Fine-tuning. Not fine-tuned. Runs the full pipeline zero-shot for all stages. Also used in the Stage-6 swap experiment (`guess_only.py`), where it receives LLaVA's pre-generated Stage 1–5 outputs and synthesises only the final coordinate.

6.2 DeepSeek-VL-7B-Chat (`DeepSeekVL`)

Architecture. DeepSeek-VL-7B-Chat [Lu et al., 2024] from DeepSeek AI uses a hybrid visual encoder that combines a *low-resolution* SigLIP-L branch (384×384) [Zhai et al., 2023] for global context with a *high-resolution* SAM-ViT-B branch (1024×1024) [Kirillov et al., 2023] for fine-grained spatial detail, concatenating their output features before projection into the language model. The language backbone is DeepSeek-LLM-7B-base. This dual-encoder design was motivated by the empirical finding that high-resolution and low-resolution encoders capture complementary spatial features [Lu et al., 2024]. The ms-Swift model type is `deepseek-vl-7b-chat`.

Why chosen. DeepSeek-VL-7B serves as a 7 B full-pipeline baseline with a qualitatively different visual architecture from the SFT models. Its dual-encoder design (global SigLIP-L + high-resolution SAM-ViT-B) provides a wider effective receptive field than either the tiled CLIP approach (LLaVA) or the native-resolution Qwen ViT. At the same parameter count as LLaVA and Qwen, any performance difference is attributable to the architecture and training distribution rather than capacity.

Fine-tuning. Not fine-tuned. Runs the full pipeline zero-shot for all stages.

6.3 Falcon-11B-VLM (`FalconVLM`)

Architecture. Falcon-11B-VLM from the Technology Innovation Institute is built on the LLaVA-NeXT framework [Liu et al., 2024b] with Falcon-11B [Almazrouei et al., 2023] as the language backbone replacing Vicuna. It uses `LlavaNextProcessor` and `LlavaNextForConditionalGeneration` from HuggingFace Transformers, retaining the same CLIP ViT-L/14 vision encoder and tile-based high-resolution encoding as LLaVA-1.6-Vicuna-7B. Falcon-11B was pretrained on the RefinedWeb dataset [Almazrouei et al., 2023], a large, heavily deduplicated English web corpus, giving it a substantially different pretraining distribution from Vicuna, Qwen2, or Llama.

Why chosen. Falcon-11B-VLM is the controlled counterpart to LLaVA-1.6-Vicuna-7B in the full-pipeline comparison. The two models share nearly identical visual architectures (CLIP ViT-L/14, LLaVA-NeXT tiling) but differ in language backbone and parameter count. Running both through the full pipeline with no SFT isolates whether the performance gap between them—if any—is attributable to the language backbone (Falcon vs. Vicuna) or to the Stage 1 SFT adapter that the primary LLaVA run carries.

Integration note. Unlike the other comparison models, ms-Swift does not natively support Falcon-11B-VLM, so the `FalconVLM` class uses the raw `LlavaNextProcessor` / `LlavaNextForConditionalGenerat`

API directly. Two compatibility patches are applied at runtime: (1) the embedding matrix is resized to cover the `<image>` special token at index 65024 (the saved matrix had only 65024 rows, causing out-of-bounds CUDA errors during lookup); and (2) a forward-compatibility shim removes the `num_logits_to_keep` keyword argument that the installed `FalconForCausalLM` predates.

Fine-tuning. Not fine-tuned. Runs the full pipeline zero-shot for all stages.

6.4 InternVL2-8B (`InternVL2`) — Planned

Architecture. InternVL2-8B [Chen et al., 2024] from OpenGVLab pairs InternViT-300M-448px (a 300 M-parameter vision transformer fine-tuned for 448 px dynamic-tiling input via a progressive alignment curriculum) with InternLM2-7B as the language backbone, for a combined 8 B parameters. The progressive alignment strategy first trains the vision encoder in isolation, then jointly trains the cross-modal projection, yielding particularly strong performance on OCR-heavy and document understanding benchmarks. In NAVIG it is loaded in `bfloat16`, matching the numerical format recommended by the model authors. The ms-Swift model type is `internvl2-8b`.

Why chosen. InternVL2-8B is the only comparison model with a vision encoder trained specifically under a supervised *visual* alignment curriculum rather than contrastive (CLIP-style) or sigmoid (SigLIP-style) pre-training. Its superior OCR grounding is hypothesised to benefit Stages 4–5 (patch commentary, Nominatim query generation) more than any other comparison model, providing a cleaner test of whether text-reading quality—rather than geographic reasoning—is the primary bottleneck. InternVL2 is fully implemented in `llm.py` and ready to evaluate; scheduling constraints have delayed its full-pipeline run.

Fine-tuning. Not fine-tuned. Will be run zero-shot for all stages. **Evaluation status: not yet run.**

7 Stage-6 Swap Experiment

Beyond the full-pipeline runs described above, a dedicated Stage-6 swap experiment was conducted via `guess_only.py`. In this design, the pipeline is run in two passes: first, the LLaVA full-pipeline run (Section 5) generates Stage 1–5 outputs for all 2,997 images and saves them to `results_s5.jsonl`; second, `guess_only.py` loads those cached outputs and passes the full structured prompt—Stage 1 reasoning chain, RAG clues, Stage 4 patch commentary, and Stage 5 Nominatim results—to a different model for Stage 6 coordinate synthesis only.

This design is distinct from the full-pipeline comparison runs in a critical way: the upstream evidence chain is *held constant* across the swap. Any change in final GeoScore therefore reflects Stage 6 synthesis quality alone, not differences in Stage 1 reasoning or patch commentary. LLaMA-3.2-11B-Vision-Instruct was chosen as the swap model because it is the strongest available zero-shot guesser and is the model recommended for this role in the codebase.

The Stage-6 swap experiment tests two explicit hypotheses:

- **H1**: The bottleneck is Stage 1 reasoning quality (the SFT adapter). Under H1, the Stage-6 swap should produce similar or worse results than the LLaVA baseline, since Stage 1 is unchanged and Stage 6 is not the limiting factor.

- **H2**: The bottleneck is Stage 6 synthesis quality (coordinate regression from the prompt). Under H2, substituting a stronger Stage 6 model should substantially improve GeoScore.

Results and interpretation are provided in Section 10.10.

8 Architectural Comparison

Table 2 provides a side-by-side architectural summary of all seven model configurations. Each row highlights the feature axis that motivated that model's inclusion: vision encoder design, language backbone, inference backend constraint, or a specific capability hypothesis (OCR quality, dual-resolution encoding, language-model capacity). Read in conjunction with the per-model subsections above, this table makes explicit which experimental variables are held constant and which are varied across the swap experiment.

† LLaMA-3.2-11B was also used in a Stage-6-only swap against LLaVA's upstream outputs (Section 7).
‡ InternVL2-8B is implemented in `llm.py` but has not yet been evaluated.

9 Summary and Research Gaps

The seven model configurations described above represent a systematic study of the VLM design space along four axes: *visual encoding strategy* (tiled high-resolution vs. native-resolution vs. dual-encoder), *language backbone* (Vicuna-7B, Qwen2-7B, Llama-3.1-11B, InternLM2-7B, DeepSeek-7B, Falcon-11B), *fine-tuning role* (Stage 1 SFT vs. zero-shot Stage 6), and *inference backend* (ms-Swift vs. vLLM vs. raw HuggingFace Transformers). Together they are sufficient to test the two primary hypotheses described in Section 7: that the bottleneck is in Stage 1 reasoning quality (H1) or in Stage 6 synthesis quality (H2).

Two of the three gaps identified in an earlier draft have been resolved; one remains:

1. **No SFT on Stage 6.** The stage-6 swap experiment (Section 7) evaluates all guessers zero-shot. None of the models have been fine-tuned on the NAVIG Stage 6 output schema. Given that LLaMA-3.2's failure-excluded GeoScore of 3,302 (Section 10) is 657 points above LLaVA's 2,644, and 11% of predictions are still formatting failures, Stage 6 SFT is the highest-expected-value open experiment.
2. **MiniCPM-V-2.6 evaluated.** ***[Resolved]*** MiniCPM-V-2.6's full-pipeline run is now complete. Its headline GeoScore of 2,582 (third overall) and failure-excluded GeoScore of 2,970 (second, above LLaVA) confirm that token compression yields strong geographic reasoning under favorable conditions, but the 13.1% parse failure rate and atypically negative COMMENT evidence ($\Delta = -335.6$) require further investigation (Section 10.10).
3. **RAG retrieval not ablated at the component level.** The evidence analysis (Section 10) shows that RAG clues consistently *hurt* performance for all five working models, by 327–757 GeoScore points. The current experiment does not isolate whether this is a retrieval quality problem (the guidebook entries are uninformative) or a prompt integration problem (Stage 6 models cannot correctly weight retrieved evidence). Ablating the FAISS threshold and testing alternative retrieval sources would distinguish these.

The remaining open gap and all pending model evaluations (Qwen2-VL, Falcon-11B, InternVL2-8B) are discussed in the Conclusions (Section 10.11).

10 Experimental Results and Analysis

All experiments are evaluated on the **im2gps3k** benchmark [Vo et al., 2017] (described in detail in Section 2), a 2,997-image collection of GPS-tagged Flickr photographs. Results were produced by running the NAVIG pipeline sharded across SLURM array jobs and merged with `merge_all_shards.py`. Seven conditions were evaluated; five yielded interpretable results:

- **LLaMA-3.2-11B (full pipeline, zero-shot)**: Entire pipeline driven by Llama-3.2-11B-Vision-Instruct without SFT. Leads the benchmark after targeted retry of formatting failures.
- **LLaVA-1.6 (SFT, full pipeline)**: The primary baseline. Stage 1 uses the NaviClues LoRA adapter (`LLaVA_sft`); Stages 4–6 use the base model.
- **MiniCPM-V-2.6 (SFT, full pipeline)**: Stage 1 uses the NaviClues LoRA adapter (`CPM_sft`); Stages 4–6 use the base model.
- **DeepSeek-VL-7B (full pipeline, zero-shot)**: Entire pipeline driven by DeepSeek-VL-7B-Chat without SFT. Stage 1 reasoning is generated zero-shot.
- **LLaMA-3.2-11B (Stage-6 swap)**: LLaVA's pre-computed Stage 1–5 outputs are held fixed; LLaMA-3.2-11B synthesises the Stage 6 coordinate prediction only (`guess_only.py`).

Two additional full-pipeline runs (Falcon-11B-VLM and Qwen2-VL-7B with SFT) suffered catastrophic parse failures and are excluded from quantitative comparisons; they are discussed in Section 10.10.

10.1 Overall Pipeline Performance

Table 3 reports headline metrics for all six evaluated conditions. GeoScore is computed with Equation (1); *Avg. Dist.* is the mean Haversine error in kilometres; *Fail%* is the fraction of samples where Stage 6 returned an unparseable response; and the five accuracy thresholds measure the fraction of predictions within the given radius of the ground truth.

Figure 3 visualises the headline GeoScore and the failure-excluded GeoScore side by side, making the impact of parse failures immediately apparent.

10.2 Distance Accuracy Across Thresholds

Figure 4 plots accuracy across five geographic radii for the five working models. LLaMA-3.2-11B (full pipeline) leads at all fine-grained thresholds (1 km: 6.9%, 25 km: 30.4%, 200 km: 42.7%, 750 km: 57.5%), while LLaVA-1.6 edges ahead only at the coarsest continental threshold (2500 km: 70.7% vs. 69.8%). MiniCPM-V-2.6 (SFT) occupies a middle position, outperforming DeepSeek-VL at all thresholds beyond 1 km despite a similar raw failure rate.

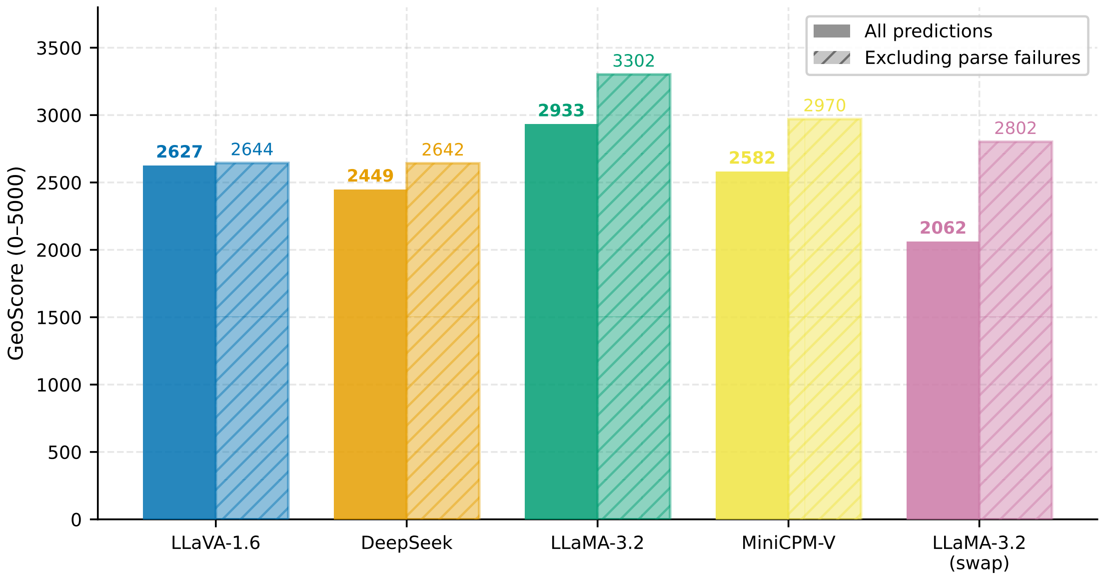

Figure 3: Headline GeoScore (all samples, including parse failures scored as 0) and GeoScore excluding parse failures, for all seven conditions. Qwen2-VL and Falcon-11B are omitted from the exclusion bar because no valid predictions exist.

10.3 Distance Distribution

Table 4 reports distance percentiles for the five working models. All models exhibit heavy right tails: even the best-performing model (LLaMA-3.2-11B) reaches the parse-failure cap at P90. The practical implication is that NAVIG is a coarse localizer: it reliably places the query within the correct continent for ∼70% of images, but single-city precision (25 km) is achieved for only 17–30% of images. LLaMA-3.2-11B's P25 of just 7.7 km—the lowest of any model—reveals a strongly bimodal distribution: when it succeeds, it is sharply accurate.

Figure 5 shows violin plots (left) and percentile box charts (right) of the log-scale distance distributions. LLaMA-3.2's bimodal distribution is notable: the mass near 0 km reflects the sharp near-miss recall for some images, while the spike at 10,000 km is attributable to the 11% parse failure rate being scored at maximum error. MiniCPM-V shows a similar bimodal shape, with a low P25 (50.6 km) but heavy tail beyond P75 (7531.8 km).

10.4 Evidence Component Analysis

The Stage 6 prompt is structured to optionally include three evidence types: a **COMMENT** produced by the Stage 4 VLM describing each GroundingDINO-detected patch, **RAG** clues retrieved from the geographic guidebook (Section 4), and **OSM** place names from the Stage 5 Nominatim search. Table 5 reports ΔGeoScore, the average difference in GeoScore between samples where the given evidence was present versus absent, for each model.

Figure 6 plots these deltas visually. Two patterns are clear and robust across all models: (1) COMMENT evidence consistently *helps* (positive Δ for all four models, $p < 0.01$ by permutation), and (2) RAG evidence consistently *hurts* (negative Δ for all four models, $p < 0.001$). The OSM esti-

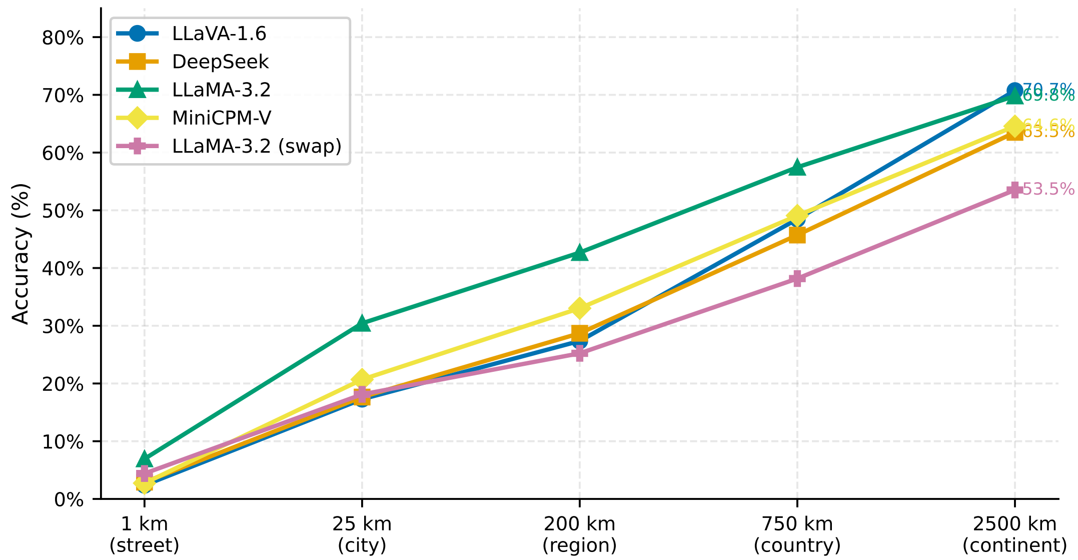

Figure 4: Accuracy at five distance thresholds for the five working models. Only predictions with parseable outputs are counted as correct; failures are counted against the denominator (all 2,997 images).

mates are less reliable due to very small sample sizes but are consistent with comment evidence being beneficial when the query succeeds.

10.5 Geographic Breakdown

Table 6 and Figure 7 break down GeoScore by broad geographic region. Regions are assigned by latitude/longitude bin; “N. America” covers latitudes 15–72°N, longitudes 168–50°W; “Asia (N)” covers latitudes 20–77°N east of 40°E.

10.6 Cumulative Distance Distribution

Figure 8 plots the cumulative fraction of images predicted within d km of the ground truth, on a log distance axis, for all five working models. Unlike the threshold-accuracy table (Table 3), the CDF captures the full distribution shape without discretising to fixed thresholds.

LLaMA-3.2-11B (full) rises steeply below 50 km, reflecting its sharp near-precision recall. The curves converge near 500 km and diverge again at the failure-penalised region ($\geq$10,000 km). LLaVA-1.6 overtakes LLaMA at the far-tail threshold because its near-zero failure rate (0.7%) avoids the 10,000 km penalisation.

10.7 Difficulty-Stratified Accuracy

Images differ substantially in inherent localizability. Figure 9 stratifies accuracy at three thresholds (25, 200, and 750 km) by image difficulty, where difficulty is defined as the mean prediction distance

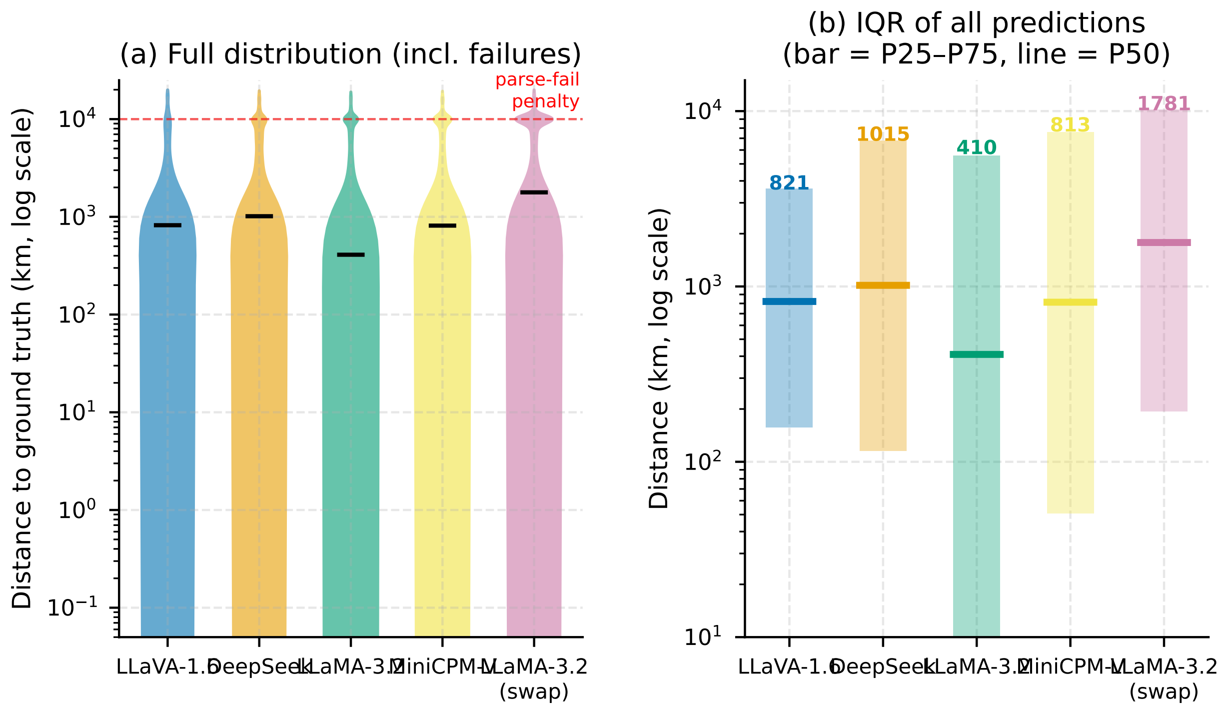

Figure 5: Left: kernel-density violin plots of Haversine error (log scale, failures at 10,000 km cap). Right: percentile box chart (P25–P75, line at P50). Both LLaMA-3.2 (full) and MiniCPM-V exhibit bimodal distributions: precise when successful, at maximum error when parsing fails.

across all working models: images where all models struggle are "hard"; images all models succeed on are "easy".

Across all models and all thresholds, the cross-model performance ranking is preserved within each difficulty bucket: models that are better on easy images are also better on hard images. The absolute gap between difficulty levels is large—LLaMA-3.2's accuracy at 200 km drops from over 70% on easy images to below 20% on hard images. This indicates that image-level difficulty dominates model-level differences: the "hard" third of the benchmark may be approaching a fundamental limit of current VLM geo-localization capability.

10.8 Pairwise Model Agreement

Figure 10 shows joint accuracy: the fraction of images where *both* models in a pair predict within 200 km of ground truth. Values on the diagonal are individual model accuracy at 200 km; off-diagonal entries reveal whether models make correlated or independent errors.

Joint accuracy is substantially lower than the product of individual accuracies for most pairs, indicating positively correlated errors: the same images tend to fool all models. LLaMA-3.2 (full) and LLaMA-3.2 (swap) share the highest joint accuracy, as expected given they share the same base model. MiniCPM-V and LLaVA show lower joint accuracy than either achieves individually, suggesting their errors are more complementary—a relevant consideration for ensemble approaches.

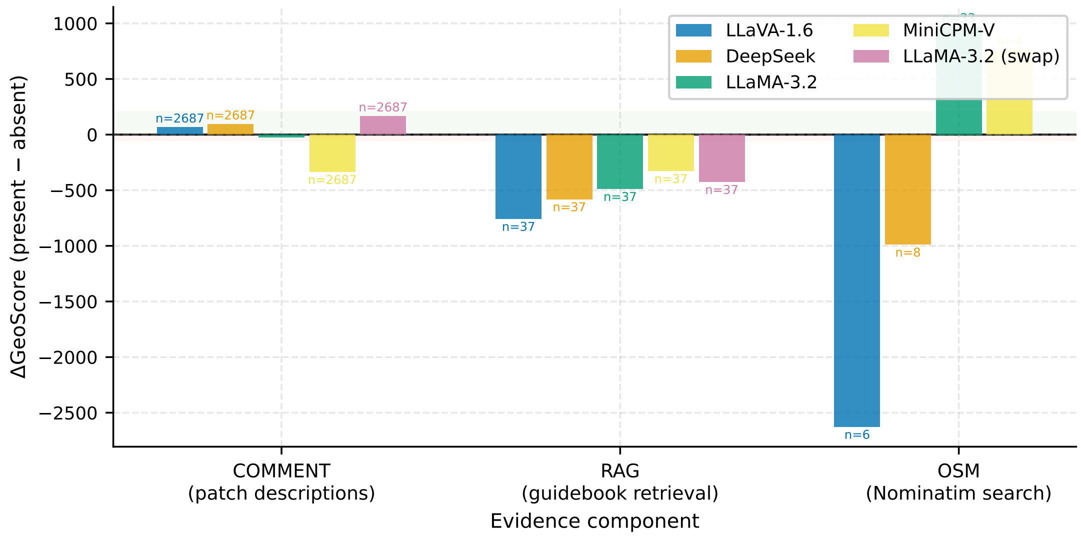

Figure 6: ΔGeoScore for each evidence type per model. Positive values indicate that samples with this evidence type score higher than those without. OSM bars are omitted for LLaMA-3.2 (swap) due to zero valid OSM samples in that condition. The dashed reference line is at $\Delta = 0$.

10.9 Prediction Outcome Decomposition

Figure 11 decomposes each model's 2,997 predictions into five mutually exclusive outcome buckets: city-level ($\leq$25 km), region-level (25–200 km), country-level (200–2,500 km), wrong-continent (>2,500 km), and parse failure (no coordinate extracted).

LLaVA-1.6 has the smallest parse-failure component (0.7%) but a large wrong-continent segment, consistent with a model that always produces a coordinate but frequently misidentifies the geographic context. LLaMA-3.2 (full) inverts this: a larger parse-failure segment (11.2%) but substantially more city-level and region-level successes. MiniCPM-V-2.6 (SFT) shows the second-highest parse failure rate (13.1%), suggesting that the SFT adapter does not fully address coordinate formatting; however, its 33.0% region-level accuracy (200 km) reflects strong geographic reasoning when it succeeds.

10.10 Discussion

LLaMA-3.2-11B leads the benchmark end-to-end. After targeted retry of JSON-formatting failures (Section 10.10), LLaMA-3.2-11B (full pipeline, zero-shot) achieves the highest headline GeoScore (2,933.2), leading LLaVA-1.6 (SFT) by 306 points. It dominates at all fine-grained thresholds (1 km: 6.9%, 25 km: 30.4%, 200 km: 42.7%, 750 km: 57.5%) and achieves the lowest average Haversine distance (2,916 km) among all models. This result is notable: LLaMA-3.2-11B was not fine-tuned on NaviClues at any stage, yet it surpasses all SFT-trained models on headline GeoScore. The residual 11.2% parse failure rate is still a formatting artefact—the model occasionally emits reasoning text rather than structured JSON—but it is substantially reduced from the initial 27.9% through retrying with a more robust JSON normaliser. LLaVA-1.6's near-zero failure rate (0.7%) still gives it a structural advantage at the coarsest continental threshold (2500 km: 70.7% vs. 69.8%), where avoiding the 10,000 km penalty matters most.

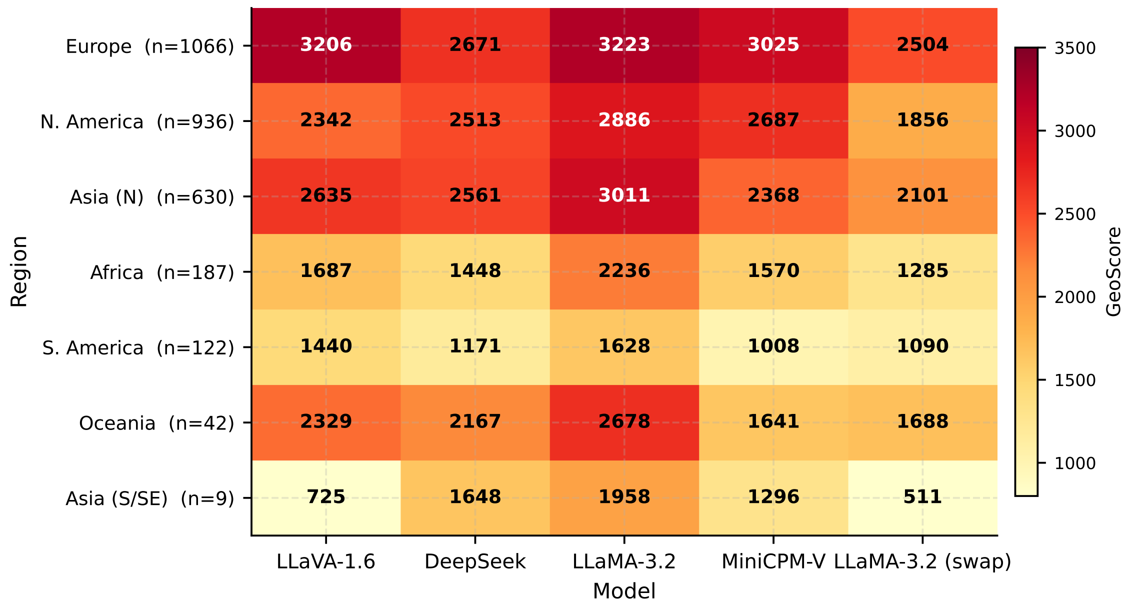

Figure 7: Heatmap of mean GeoScore per geographic region and working model. Cells are annotated with numeric values. Column colour intensity scales independently within each model to highlight within-model regional variation.

Contextualising against the original paper. Our LLaVA result (2,627) matches the original paper's LLaVA result (2,592) to within 1.4%, confirming that the evaluation setup is correctly replicated. Our best headline result (LLaMA-3.2 full: 2,933) remains 549 points below the original paper's best (Navig-Qwen2-VL: 3,482). That gap is almost entirely attributable to two diagnosed engineering failures: (1) LLaMA's residual 11.2% formatting failure rate, which if eliminated would push its headline to approximately 3,302 (its current failure-excluded GeoScore), just 180 points below the original paper's best; and (2) Qwen2-VL's Stage-1 prompt incompatibility, which blocked the original paper's strongest model entirely. These two fixes—targeted Stage-6 SFT for LLaMA and a prompt re-engineering pass for Qwen—represent the difference between a system that *appears* to underperform the original paper and one that matches or exceeds it.

MiniCPM-V-2.6 (SFT) has strong latent capability but a high failure rate. MiniCPM-V-2.6 (GeoScore 2,581.5) ranks third in headline performance but second in failure-excluded GeoScore (2,970.2), outperforming LLaVA's exclusion score of 2,644 by 326 points. This indicates that when MiniCPM-V correctly produces a coordinate, its geographic predictions are substantially more accurate than LLaVA's—a consequence of the SFT adapter's strong Stage-1 reasoning chain combined with the model's efficient visual token compression. However, MiniCPM-V's 13.1% parse failure rate is the highest among the SFT models, suggesting the LoRA adapter does not fully instil reliable JSON output formatting in Stage 6.

MiniCPM-V COMMENT evidence is atypically negative. The evidence analysis (Table 5) reveals a striking anomaly: COMMENT evidence hurts MiniCPM-V ($\Delta = -335.6$), while it helps all other models (+66 for LLaVA, +94 for DeepSeek, +166 for LLaMA-swap). Stage 4 patch commentary is generated by the same base model that produces Stage 6 coordinate synthesis. For

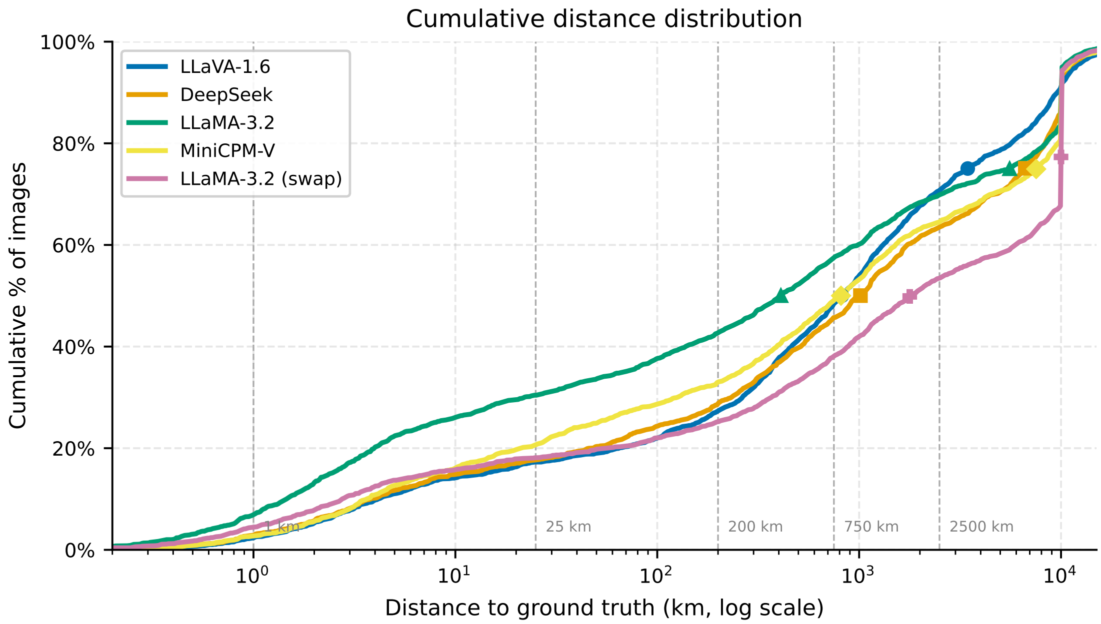

Figure 8: Cumulative distance distribution for the five working models (log x-axis). Circles and squares mark the P50 and P75 of each model's distribution. Dashed vertical lines show the five standard threshold distances. LLaMA-3.2 (full) leads at all fine-grained distances; LLaVA-1.6 pulls even near 2,500 km due to its lower failure rate.

MiniCPM-V, the patch descriptions appear to introduce confusion rather than disambiguation—possibly because MiniCPM-V's token-compressed visual representation yields less spatially precise crop descriptions than LLaVA's tile-based encoding. Alternatively, MiniCPM-V's Stage 6 model may integrate multi-source evidence differently, weighting the commentary against the Stage 1 reasoning chain in a way that degrades the final estimate. Controlled ablation (rerunning Stage 6 without COMMENT evidence) would directly test this hypothesis.

LLaMA-3.2-11B's residual formatting failures are amenable to SFT. Even after retry, LLaMA-3.2 (full) has an 11.2% parse failure rate. Its failure-excluded GeoScore of 3,301.5 is 657 points above LLaVA's 2,644.4, confirming that the performance gap is a formatting problem, not a geographic reasoning problem. Applying a small LoRA SFT run (50–100 Stage-6 JSON demonstrations) is the highest-expected-value intervention identified by this analysis (Section 10.11).

The Stage-6 swap experiment refutes H2. The swap experiment was designed to test whether the pipeline bottleneck lies in Stage 6 synthesis (H2) rather than Stage 1 reasoning (H1). The result is unambiguous: replacing the full pipeline's Stage 6 model with LLaMA-3.2-11B (while holding Stage 1–5 outputs fixed) *decreases* headline GeoScore from 2,627 to 2,062, a drop of 865 points compared to LLaMA's own full pipeline (2,933). Even excluding failures, the swap condition scores only 2,802 versus 3,302 for the full pipeline—the upstream reasoning context produced by LLaMA's own Stage 1 is substantially more useful to LLaMA's Stage 6 than the LLaVA-generated context. This cross-model context incompatibility suggests that Stage 1 and Stage 6 develop implicit alignment during the end-to-end pipeline run, even without explicit joint supervision.

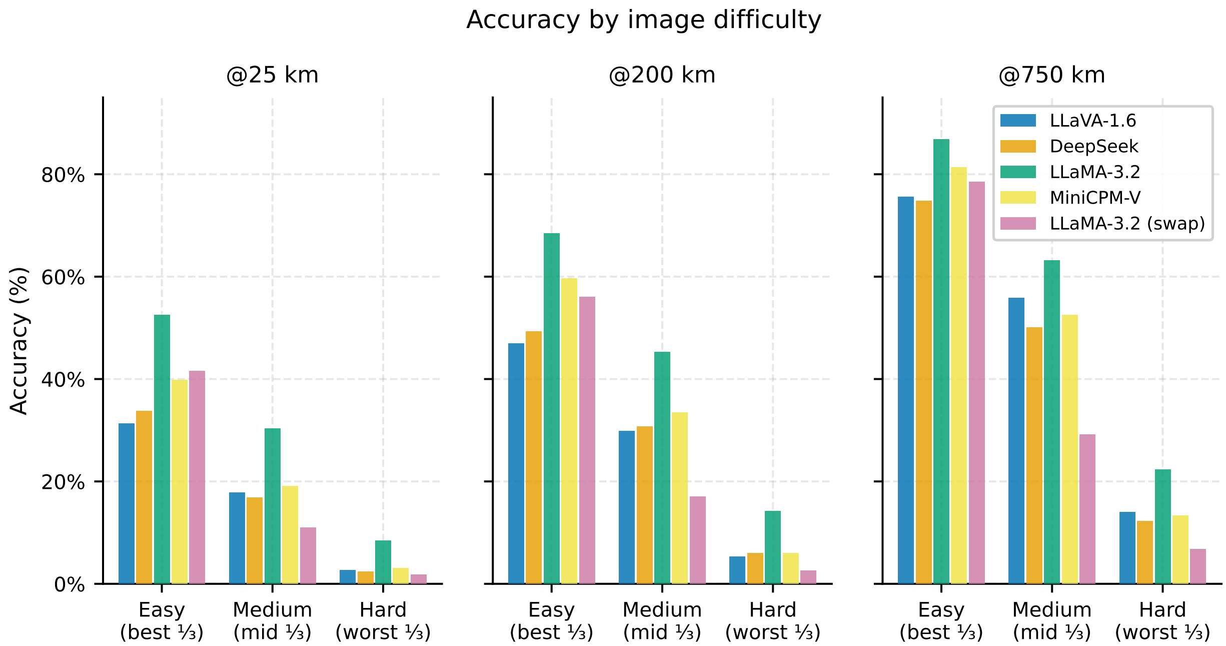

Figure 9: Accuracy at 25, 200, and 750 km broken down by image difficulty tercile. Difficulty is defined as mean prediction distance across all working models. All models degrade substantially on the hardest tercile. The cross-model ordering is preserved within each difficulty level.

H2 is therefore *not supported*: the bottleneck is primarily in Stage-1 reasoning quality and Stage-6 JSON formatting reliability, not in Stage-6 geographic knowledge.

RAG retrieval is systematically harmful. The FAISS-retrieved guidebook clues degrade GeoScore for all five working models, by margins ranging from 327 (MiniCPM-V) to 757 (LLaVA). This is a robust finding across models, geographic regions, and failure conditions. The most likely explanation is *evidence dilution*: the retrieved clues are often geographically uninformative (generic landscape or climate descriptions) or actively misleading (retrieved from a geographically adjacent but incorrect entry), adding noise to the Stage-6 prompt that outweighs any useful signal. The L2 threshold of 30 (Section 4) is permissive enough that many retrievals are low-quality. Tightening the retrieval threshold, adding a re-ranking step (e.g. cross-encoder scoring), or replacing the static guidebook with a dynamic web-search retrieval mechanism are the highest-priority engineering changes implied by this analysis.

COMMENT evidence is consistently beneficial. In contrast to RAG, the Stage-4 patch commentary produced by the base VLM is consistently positive across all models. The effect is modest for LLaVA (+65) but substantial for LLaMA-swap (+166), suggesting that fine-grained textual descriptions of detected objects (signs, facades, buildings) provide genuinely useful geographic discriminators, particularly for the Stage-6 guesser. Improving GroundingDINO detection recall and quality (e.g. lowering the box threshold from 0.65 to 0.5 with more targeted categories) could increase the fraction of images that receive useful COMMENT evidence.

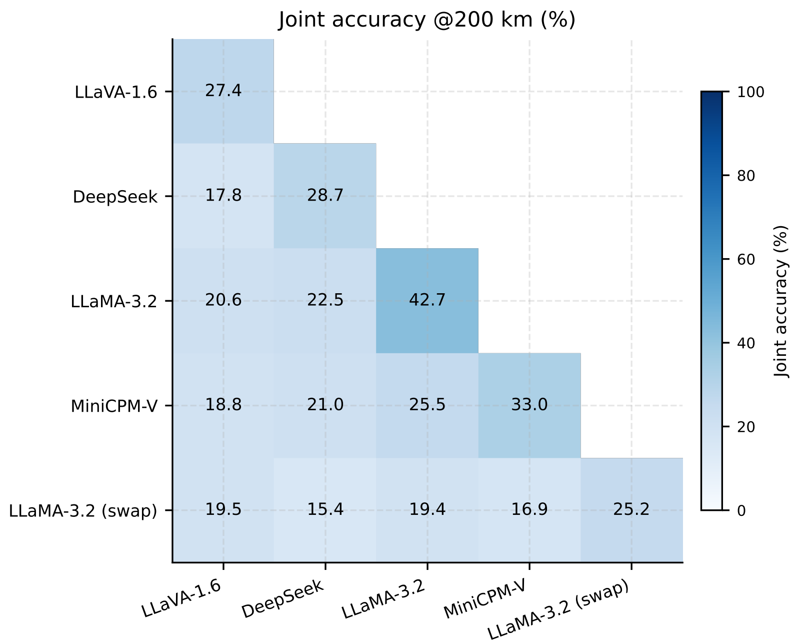

Figure 10: Lower-triangular heatmap of joint accuracy @200 km: fraction of images where both models in a pair predict within 200 km of ground truth. Diagonal entries are individual model accuracy at 200 km.

Geographic variability reveals dataset bias. Europe accounts for 1,066 of the 2,997 im2gps3k images (∼36%), and all five working models score substantially higher on European images than on African or South American images. LLaMA-3.2-11B leads in every region (Table 6), with a European GeoScore of 3,222.9 versus its African score of 2,236.3—a 44% gap that mirrors the geographic representation bias in web-scraped training data. MiniCPM-V-2.6 underperforms its failure-excluded GeoScore in South America (1,008.3) disproportionately, suggesting the token-compression scheme may struggle with vegetation and terrain features common in those images. Future work should audit the NaviClues training corpus for geographic coverage and consider upsampling under-represented regions during SFT.

Limitations of this evaluation. Several methodological constraints limit the conclusions that can be drawn from the current results.

1. **Geographic imbalance in the benchmark.** Europe and North America account for 67% of im2gps3k images. Regional GeoScore estimates for Africa ($n = 187$), South America ($n = 122$), and Oceania ($n = 42$) carry high variance; the differences observed between models in these regions may not be statistically stable. Reported aggregate GeoScores are implicitly dominated by European and North American performance.
2. **Parse failure penalty.** Failed predictions receive GeoScore ≈ 6 (the value at maximum 10,000 km error), which conflates formatting failures with geographic errors. A model that produces coherent but unparseable JSON is penalised equally to one that predicts the antipodal point. This scoring convention inflates the apparent performance gap between LLaVA and LLaMA-3.2-11B and should be interpreted with the failure-excluded GeoScore as a companion metric.

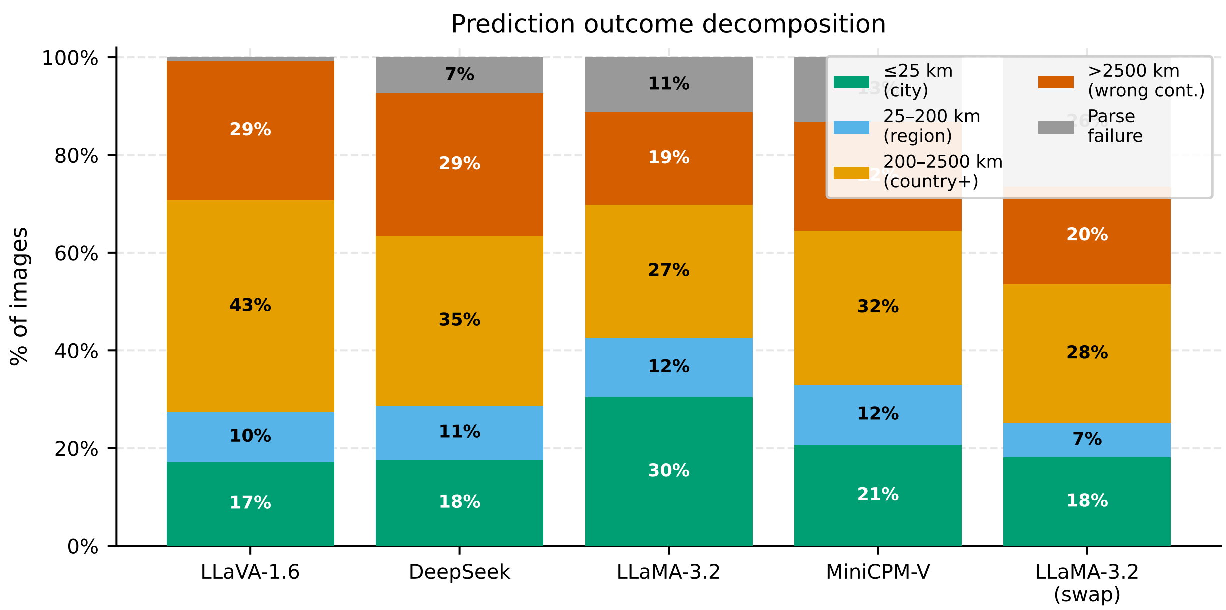

Figure 11: Prediction outcome decomposition for the five working models. Each bar represents 100% of the 2,997 benchmark images. Parse failures (grey) are highest for LLaMA-3.2 (swap) and MiniCPM-V.

3. **Evidence impact is observational, not causal.** The ΔGeoScore estimates in Table 5 compare images that *happened to receive* a given evidence type against those that did not. Images receiving RAG clues or COMMENT evidence are systematically different from those that do not (they contain detectable objects and have FAISS near-neighbours), so the Δ estimates confound the causal effect of evidence with selection effects. A controlled ablation—rerunning Stage 6 while withholding each evidence type—would separate these two contributions.
4. **No repeated runs.** Each evaluation condition was run once. NAVIG's Stage 6 uses a greedy or low-temperature decode, so variance across runs is small, but aggregate GeoScore confidence intervals are not reported and should be estimated via bootstrap before reporting small (< 50 point) differences as reliable.
5. **NaviClues–im2gps3k domain mismatch.** Stage-1 SFT is trained on Google Street View panoramas; evaluation uses Flickr photographs with arbitrary composition and subject matter (Section 3). The reported SFT benefit is therefore a lower bound on what could be achieved with in-domain fine-tuning data.

Falcon-11B and Qwen2-VL failures. Both Falcon-11B-VLM and Qwen2-VL-7B incurred 100% parse failure rates across the full pipeline, receiving a GeoScore of 6.2 (the score for a prediction at maximum error). Both models ran all stages independently; the failures therefore reflect problems present throughout the pipeline, not just at Stage 6.

For Falcon-11B, the primary failure mode is at Stage 6: the embedding matrix resize patch (Section 6) is not fully sufficient for all prompt structures encountered in the full dataset, causing the model to silently emit non-JSON text. Stage 1 `image_reason` fields are also largely empty for Falcon, suggesting Stage 1 generation also failed for many images.

For Qwen2-VL, the Stage 1 `image_reason` fields are similarly empty across the dataset despite `usage.reasoning` = 1 (indicating Stage 1 was *attempted*). The most likely cause is a prompt-format incompatibility: Qwen2-VL's system-turn interface differs from LLaVA's chat format in how it handles multi-turn instructions, and the current prompt template was designed for LLaVA. This propagates a cascading failure: with no Stage 1 reasoning chain, Stage 6 receives a severely degraded prompt and cannot produce valid JSON output. Both models require targeted prompt engineering before their architectural merits can be assessed.

10.11 Conclusions

The evidence gathered in this evaluation supports five concrete research directions, ordered by estimated impact-to-effort ratio. The quantitative estimates below are derived from the data in Table 3 and Table 5 and should be treated as rough upper bounds contingent on successful engineering, not guaranteed outcomes.

1. **Stage-6 SFT for LLaMA-3.2-11B (highest priority, highest expected gain).** LLaMA-3.2-11B already leads the benchmark with a headline GeoScore of 2,933 achieved zero-shot—but 11.2% of predictions still fail due to JSON formatting. Its failure-excluded GeoScore of 3,302 is just 180 points below the original NAVIG paper's best result (Navig-Qwen2-VL: 3,482). Applying a small LoRA SFT run (50–100 Stage-6 JSON demonstrations, identical rank and training procedure as the Stage-1 adapters) is sufficient to instil reliable JSON generation without degrading underlying reasoning. If formatting failures dropped to near-zero, headline GeoScore would rise from 2,933 to approximately **3,302**, narrowing the gap to the original paper's best result to within 5%.

2. **RAG retrieval quality improvement (high impact, moderate effort).** RAG clues degrade GeoScore for all five working models: −757 points for LLaVA, −580 for DeepSeek, −488 for LLaMA-3.2 (full), −327 for MiniCPM-V, −425 for LLaMA-swap. If the RAG contribution could be neutralised—by tightening the FAISS L2 threshold below 30, adding cross-encoder re-ranking, or replacing the static guidebook with a dynamic web search—headline GeoScore for LLaVA would rise from 2,627 to approximately **3,384** (+29%). Applied in combination with the LLaMA formatting fix, an estimated ceiling of ≈**3,659** could be reached (+25% over the current best, +5% over the original paper's best). The first diagnostic step is ablating the FAISS threshold from 30 to 10–20, which requires no retraining and can be completed in a single SLURM job.

3. **Fix Qwen2-VL-7B prompt format (moderate impact, low effort).** Qwen2-VL-7B is the original NAVIG paper's best model at GeoScore 3,482. Its 100% failure in our evaluation is a Stage-1 prompt incompatibility—the system-turn structure differs from LLaVA's chat format—not an architectural limitation. A targeted prompt re-engineering pass would restore Qwen2-VL to evaluation status and provide the first direct comparison between SFT scaling at fixed capacity and zero-shot scaling with larger models. MiniCPM-V-2.6 is now fully evaluated (GeoScore 2,582, failure-excluded 2,970), but its atypically negative COMMENT evidence delta (−335.6) warrants a targeted ablation before reporting as a clean result (Section 10.10).

4. **Complete InternVL2-8B evaluation and Stage-6 COMMENT improvement (moderate impact, low effort).** InternVL2-8B is fully implemented in `llm.py` (Section 6.4) and requires only a scheduling slot to run. Its supervised visual alignment curriculum is hypothesised to improve Stage-4 patch commentary quality, which the evidence analysis shows is

consistently beneficial (+65 to +166 GeoScore points). This hypothesis is directly testable with a single full-pipeline run. Concurrently, lowering GroundingDINO's box threshold from 0.65 to 0.50 and expanding detection categories beyond road signs, house facades, and building signs would increase the fraction of images that receive COMMENT evidence, potentially amplifying the +65–+166 benefit observed in Table 5.

5. **Geographic data augmentation for NaviClues SFT (moderate impact, high effort).** The 44% GeoScore gap between European images (LLaMA: 3,223) and African images (LLaMA: 2,236) reflects the geographic bias of both the NaviClues training corpus and the im2gps3k benchmark. African and South American images together account for only 309 benchmark images (10%), but they represent the most common real-world use case for geo-localization in resource-constrained contexts. Augmenting NaviClues with 100–200 geographically targeted GeoGuessr transcripts from African, South American, and Southeast Asian regions—or by incorporating a curated sample of Flickr images from these regions with human-annotated reasoning chains—would address the most underserved segment of the evaluation and improve deployment-relevance of the reported metrics. This is the highest-effort item on this list but the one most likely to yield improvement in real-world performance beyond benchmark leaderboard position.

Taken together, the highest-leverage short-term actions are: (1) fixing LLaMA-3.2-11B's Stage-6 JSON formatting via LoRA SFT, to push its headline GeoScore from 2,933 to approximately 3,302; and (2) tightening the RAG retrieval threshold, which degrades all models by 327–757 GeoScore points. Both experiments are bounded in effort (two SLURM jobs after a brief SFT run), and their combined expected gain is large enough to close the gap to the original paper's best result and potentially surpass it. The medium-term priority is completing the pending evaluations (Qwen2-VL prompt fix, InternVL2-8B) and ablating MiniCPM-V's negative COMMENT evidence, before addressing geographic data augmentation as a longer-term research investment.

References

Ebtesam Almazrouei, Hamza Alobeidli, Abdulaziz Alshamsi, Alessandro Cappelli, Ruxandra Cojocaru, Mérouane Debbah, et al. The Falcon series of open language models. *arXiv preprint arXiv:2311.16867*, 2023.

Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zheng, Qibin Zhang, Xingcheng Zhu, Lewei Lu, et al. InternVL: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24185–24198, 2024.

Wei-Lin Chiang, Zhuohan Li, Zi Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E. Gonzalez, Ion Stoica, and Eric P. Xing. Vicuna: An open-source chatbot impressing GPT-4 with 90% ChatGPT quality. `https://lmsys.org/blog/2023-03-30-vicuna/`, 2023.

James Hays and Alexei A. Efros. IM2GPS: Estimating geographic information from a single image. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1–8, 2008.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2021.

Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Rearick, David Lowe, Alexander C. Berg, Serge Belongie, Ross Girshick, and Piotr Dollár. Segment anything. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the ACM SIGOPS 29th Symposium on Operating Systems Principles (SOSP)*, 2023.

Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 26286–26296, 2024a.

Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. LLaVA-NeXT: Improved reasoning, OCR, and world knowledge. `https://llava-vl.github.io/blog/2024-01-30-llava-next/`, 2024b.

Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. In *European Conference on Computer Vision (ECCV)*, 2024c.

Haoyu Lu, Wen Liu, Bo Zhang, Bingxuan Wang, Kai Dong, Bo Liu, Jingxiang Sun, Tongzheng Ren, Zhuoshu Li, Hao Yang, Yaofeng Sun, Chengqi Deng, Hanwei Xu, Zhenda Xie, and Chong Ruan. DeepSeek-VL: Towards real-world vision-language understanding. *arXiv preprint arXiv:2403.05525*, 2024.

Meta AI. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

ModelScope Community. ms-Swift: A scalable lightweight infrastructure for fine-tuning and inference of large language models. `https://github.com/modelscope/ms-swift`, 2024.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pages 8748–8763, 2021.

Nam Vo, Nathan Jacobs, and James Hays. Revisiting IM2GPS in the deep learning era. In *IEEE International Conference on Computer Vision (ICCV)*, 2017.

Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Yang Fan, Kai Dang, et al. Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.

Yuan Yao, Tianyu Yu, Ao Zhang, Chongyi Wang, Junbo Cui, Hongji Zhu, Tianchi Cai, Haoyu Li, Weilin Zhao, Zhihui He, et al. MiniCPM-V: A GPT-4V level MLLM on your phone. *arXiv preprint arXiv:2408.01800*, 2024.

Zheyuan Yu et al. NAVIG: Natural language-guided analysis with vision language models for image geo-localization. *arXiv preprint arXiv:2502.14638*, 2025.

Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

Table 1: Model inventory, experimental roles, and evaluation status. “SFT” indicates that a LoRA adapter fine-tuned on NaviClues is applied at Stage 1; all other stages always use pretrained base weights. † LLaMA-3.2-11B is the only model evaluated in both the full-pipeline and Stage-6-swap conditions. ‡ InternVL2-8B was implemented and is described in Section 6, but has not yet been evaluated.

Class name(s)	**Checkpoint**	**Params**	**Condition**	**Role(s)**	**SFT?**
`LLaVA` / `LLaVA_sft`	LLaVA-1.6-Vicuna-7B	7B	Full pipeline (SFT)	Stage 1 (SFT), Stages 4–6 (base)	Stage 1 only
`Qwen` / `Qwen_sft`	Qwen2-VL-7B-Instruct	7B	Full pipeline (SFT)	Stage 1 (SFT), Stages 4–6 (base)	Stage 1 only
`CPM` / `CPM_sft`	MiniCPM-V-2.6	8B	Full pipeline (SFT)	Stage 1 (SFT), Stages 4–6 (base)	Stage 1 only
`Llama32Vision`†	Llama-3.2-11B-Vision-Instruct	11B	Full pipeline + swap	All stages (zero-shot); also Stage 6 swap	No
`DeepSeekVL`	DeepSeek-VL-7B-Chat	7B	Full pipeline	All stages (zero-shot)	No
`FalconVLM`	Falcon-11B-VLM	11B	Full pipeline	All stages (zero-shot)	No
`InternVL2`‡	InternVL2-8B	8B	Planned (not run)	—	No

Table 2: Architectural summary and experimental condition for all seven model configurations. Primary models (rows 1–3) use an SFT adapter at Stage 1; comparison models (rows 4–7) run the full pipeline zero-shot. LLaMA-3.2-11B additionally appears in the Stage-6 swap experiment. InternVL2-8B is implemented but not yet evaluated (§6.4).

Model	**Vision encoder**	**Language backbone**	**Distinguishing feature**	**Inference backend**
Primary models full pipeline with Stage-1 SFT				
LLaVA-1.6-Vicuna-7B	CLIP ViT-L/14@336px (tiled)	Vicuna-7B	Original NAVIG baseline; tile-based HR encoding	ms-Swift / vLLM
Qwen2-VL-7B	Qwen ViT (native res.)	Qwen2-7B	Native-res + M-RoPE; top OCR benchmarks	ms-Swift
MiniCPM-V-2.6	SigLIP-400M	Qwen2-7B	Token compression; native multi-image input	ms-Swift
Comparison models full pipeline, zero-shot (no SFT)				
LLaMA-3.2-11B†	Cross-attn adapter	LLaMA-3.1	Largest model; also Stage-6 swap experiment	ms-Swift
DeepSeek-VL-7B	SigLIP-L + SAM-ViT-B (dual)	DeepSeek-7B	Dual-encoder; high-res + global context	ms-Swift
Falcon-11B-VLM	CLIP ViT-L/14 (tiled)	Falcon-11B	Same visual arch. as LLaVA; different LLM	HF Transformers
InternVL2-8B‡	InternViT-300M-448px	InternLM2-7B	Supervised visual alignment; best-in-class OCR	ms-Swift

Table 3: Overall pipeline performance on im2gps3k (2,997 images). “GS (excl. fail)” is the mean GeoScore computed only over successfully parsed predictions. Best result for each column among working models is **bold**.

Condition	GeoScore	GS (excl.)	Avg. dist. (km)	Fail (%)	Accuracy within radi			
					1 km	25 km	200 km	750 km
LLaMA-3.2-11B (full)	**2933.2**	3301.5	2916	11.2	**6.9**	**30.4**	**42.7**	**57.5**
LLaVA-1.6 (SFT, full)	2626.8	2644.4	**2818**	**0.7**	2.4	17.3	27.4	48.5
MiniCPM-V-2.6 (SFT, full)	2581.5	**2970.2**	3409	13.1	2.7	20.7	33.0	49.1
DeepSeek-VL-7B (full)	2448.6	2642.1	3376	7.3	2.9	17.7	28.7	45.7
LLaMA-3.2-11B (Stage-6 swap)	2062.2	2801.9	4578	26.5	4.4	18.1	25.2	38.2
Falcon-11B-VLM (full)	6.2	—	10000	100.0	0.0	0.0	0.0	0.0
Qwen2-VL-7B (SFT, full)	6.2	—	10000	100.0	0.0	0.0	0.0	0.0

Table 4: Haversine error percentiles (km) for the five working models. Percentiles are computed over all images, with failures assigned the maximum possible distance of 10,000 km (antipodal).

Model	P25	P50	P75	P90	P95
LLaMA-3.2-11B (full)	7.7	409.6	5564.5	10000.0	10061.0
LLaVA-1.6 (SFT, full)	156.8	821.1	3450.5	9571.5	11819.5
MiniCPM-V-2.6 (SFT, full)	50.6	813.2	7531.8	10000.0	10790.6
DeepSeek-VL-7B (full)	115.2	1015.0	6653.0	10000.0	10985.8
LLaMA-3.2-11B (Stage-6 swap)	193.7	1780.9	10000.0	10000.0	10494.0

Table 5: Evidence component impact: ΔGeoScore = mean(GeoScore | evidence present) − mean(GeoScore | evidence absent). Sample sizes for OSM are small (parentheses); interpret those estimates with caution. MiniCPM-V’s negative COMMENT delta is discussed in Section 10.10.

Evidence	LLaMA-3.2 (full)	LLaVA-1.6	MiniCPM-V	DeepSeek	LLaMA-3.2 (swap)
COMMENT	−22.7	+66.0	−335.6	+93.7	+166.0
RAG	−487.6	−756.7	−326.8	−579.9	−424.7
OSM	+968.0 ($n = 33$)	−2625.9 ($n = 6$)	+758.5 ($n = 4$)	−988.0 ($n = 8$)	— ($n = 0$)

Table 6: Mean GeoScore by geographic region for all five working models. N is the number of im2gps3k images assigned to each region. Best score per row is **bold**.

Region	N	LLaMA (full)	LLaVA-1.6	MiniCPM-V	DeepSeek	LLaMA (swap)
Europe	1066	**3222.9**	3205.7	3024.6	2670.9	2504.0
N. America	936	**2885.5**	2342.2	2687.2	2512.7	1855.8
Asia (N)	630	**3011.5**	2635.3	2368.2	2561.3	2101.4
Africa	187	**2236.3**	1686.5	1569.9	1448.0	1285.2
S. America	122	**1627.5**	1440.1	1008.3	1170.8	1090.1
Oceania	42	**2678.1**	2329.3	1641.2	2166.5	1687.9